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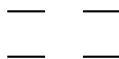
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1 Orbitals And Their Symmetry

We use the following ordering of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{cc} b_{1u} & \text{---} & \text{---} & a_u \\ b_{2g} & \text{---} & \text{---} & b_{3g} \end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:



where the character of the orbital is defined by its position.

Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital ordering within the group of monomer orbitals follows the above mentioned definition.

The transformation of monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

$$\begin{aligned} a_u^l &= +b_{1g} + a_u \\ b_{1u}^l &= +a_g + b_{1u} \\ b_{2g}^l &= +b_{3u} + b_{2g} \\ b_{3g}^l &= +b_{2u} + b_{3g} \\ a_u^r &= -b_{1g} + a_u \\ b_{1u}^r &= -a_g + b_{1u} \\ b_{2g}^r &= -b_{3u} + b_{2g} \\ b_{3g}^r &= -b_{2u} + b_{3g} \end{aligned}$$

2 Monomer States and Configurations

$$\begin{aligned}
S &= + \begin{pmatrix} \overline{} & \overline{} \\ \downarrow\uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_{3g}^l)^2 (b_{2g}^l)^2 (b_{1u}^l)^0 (a_u^l)^0}_{a_g} \right| \\
Q &= + \begin{pmatrix} \uparrow & \uparrow \\ \uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_{3g}^l)^1 (b_{2g}^l)^1 (b_{1u}^l)^1 (a_u^l)^1}_{a_g} \right| \\
i^3 b_{2u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \overline{} & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_{3g}^l)^1 (b_{2g}^l)^2 (b_{1u}^l)^1 (a_u^l)^0}_{b_{2u}} \right| - \left| \underbrace{(b_{3g}^l)^2 (b_{2g}^l)^1 (b_{1u}^l)^0 (a_u^l)^1}_{b_{2u}} \right| \\
e^3 b_{2u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \overline{} & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_{3g}^l)^1 (b_{2g}^l)^2 (b_{1u}^l)^1 (a_u^l)^0}_{b_{2u}} \right| + \left| \underbrace{(b_{3g}^l)^2 (b_{2g}^l)^1 (b_{1u}^l)^0 (a_u^l)^1}_{b_{2u}} \right| \\
e^3 b_{3u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \overline{} & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_{3g}^l)^2 (b_{2g}^l)^1 (b_{1u}^l)^1 (a_u^l)^0}_{b_{3u}} \right| - \left| \underbrace{(b_{3g}^l)^1 (b_{2g}^l)^2 (b_{1u}^l)^0 (a_u^l)^1}_{b_{3u}} \right| \\
i^3 b_{3u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \overline{} & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_{3g}^l)^2 (b_{2g}^l)^1 (b_{1u}^l)^1 (a_u^l)^0}_{b_{3u}} \right| + \left| \underbrace{(b_{3g}^l)^1 (b_{2g}^l)^2 (b_{1u}^l)^0 (a_u^l)^1}_{b_{3u}} \right|
\end{aligned}$$

3 Linear Combinations

3.1 $S * Q + Q * S$

$$S * Q + Q * S$$

$$+ \begin{pmatrix} \overline{\uparrow\uparrow} & \overline{\downarrow\uparrow} & \uparrow\uparrow & \uparrow\uparrow \\ \uparrow\uparrow & \downarrow\uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow\uparrow & \uparrow\uparrow & \overline{\downarrow\uparrow} & \overline{\downarrow\uparrow} \\ \uparrow\uparrow & \uparrow\uparrow & \downarrow\uparrow & \downarrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{3g}^l)^2 (b_{2g}^l)^2 (b_{1u}^l)^0 (a_u^l)^0}_{a_g^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1 (b_{2g}^r)^1 (b_{1u}^r)^1 (a_u^r)^1}_{a_g^r} \right| + \left| \underbrace{(b_{3g}^l)^1 (b_{2g}^l)^1 (b_{1u}^l)^1 (a_u^l)^1}_{a_g^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2 (b_{2g}^r)^2 (b_{1u}^r)^0 (a_u^r)^0}_{a_g^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+b_{2u} + b_{3g})^2 (+b_{3u} + b_{2g})^2 (+a_g + b_{1u})^0 (+b_{1g} + a_u)^0}_{a_g^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^1 (-b_{3u} + b_{2g})^1 (-a_g + b_{1u})^1 (-b_{1g} + a_u)^1}_{a_g^r} \right|$$

$$+ \left| \underbrace{(+b_{2u} + b_{3g})^1 (+b_{3u} + b_{2g})^1 (+a_g + b_{1u})^1 (+b_{1g} + a_u)^1}_{a_g^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^2 (-b_{3u} + b_{2g})^2 (-a_g + b_{1u})^0 (-b_{1g} + a_u)^0}_{a_g^r} \right|$$

multiplied out dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}
\end{array}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \right| \\
& +2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \right| \\
& +2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \right| \\
& +2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \right|
\end{aligned}$$

3.2 $S * Q - Q * S$

$$\begin{aligned}
& S * Q - Q * S \\
& + \left(\begin{array}{cc|cc} \text{---} & \text{---} & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & \uparrow & \text{---} & \text{---} \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{array} \right)
\end{aligned}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^0}_{a_g^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^1}_{a_g^r} \right| - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^1}_{a_g^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^0}_{a_g^r} \right|$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \left| \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^0}_{a_g^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^1}_{a_g^r} \right| \\
& - \left| \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^1}_{a_g^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^0}_{a_g^r} \right|
\end{aligned}$$

multiplied out dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}
\end{array}$$

summed up:

$$\begin{aligned}
& -2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \right| \\
& -2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \right| \\
& -2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \right| \\
& -2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \right|
\end{aligned}$$

3.3 $i^3 b_{2u} * i^3 b_{2u}$

$$i^3 b_{2u} * i^3 b_{2u}$$

$$+ \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right)$$

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \left| \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \right|
\end{aligned}$$

multiplied out dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g}
\end{array}$$

summed up:

$$\begin{aligned}
& + \left| \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \right| + \left| \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g} \right|
\end{aligned}$$

3.4 $i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$

$$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$$

$$\begin{aligned}
& + \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) \\
& - \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{array} \right) - \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{array} \right) - \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{array} \right)
\end{aligned}$$

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

summed up:

$$+2 \cdot \left| \underbrace{(b_{2u})^1 (b_{3g})^1 (b_{3u})^2 (b_{2g})^2 (a_g)^1 (b_{1u})^1 (b_{1g})^0 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(b_{2u})^2 (b_{3g})^2 (b_{3u})^1 (b_{2g})^1 (a_g)^0 (b_{1u})^0 (b_{1g})^1 (a_u)^1}_{a_g} \right|$$

3.5 $i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$

$$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{3g}^l)^1 (b_{2g}^l)^2 (b_{1u}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1 (b_{2g}^r)^2 (b_{1u}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{3g}^l)^1 (b_{2g}^l)^2 (b_{1u}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1 (b_{2g}^r)^2 (b_{1u}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{3g}^l)^1 (b_{2g}^l)^2 (b_{1u}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2 (b_{2g}^r)^1 (b_{1u}^r)^0 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{3g}^l)^2 (b_{2g}^l)^1 (b_{1u}^l)^0 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1 (b_{2g}^r)^2 (b_{1u}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{3g}^l)^2 (b_{2g}^l)^1 (b_{1u}^l)^0 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1 (b_{2g}^r)^2 (b_{1u}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{3g}^l)^1 (b_{2g}^l)^2 (b_{1u}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2 (b_{2g}^r)^1 (b_{1u}^r)^0 (a_u^r)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{3g}^l)^2 (b_{2g}^l)^1 (b_{1u}^l)^0 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2 (b_{2g}^r)^1 (b_{1u}^r)^0 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{3g}^l)^2 (b_{2g}^l)^1 (b_{1u}^l)^0 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2 (b_{2g}^r)^1 (b_{1u}^r)^0 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

summed up:

$$\begin{aligned}
& +4 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \right| \\
& +4 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \right| \\
& -4 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \right| \\
& -4 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \right|
\end{aligned}$$

3.6 $i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$

$$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

$$\begin{array}{lcl}
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} & - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
- & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} & + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
- & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} & + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
- & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} & + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} & + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
- & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} & - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} & + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
- & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} & - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
- & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} & + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
- & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} & + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
- & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} & + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} & - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& +2 \cdot \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} - 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
& -2 \cdot \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} + 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
& +2 \cdot \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} - 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
& -2 \cdot \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} + 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}
\end{aligned}$$

3.7 $i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$

$$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

monomer ci vectors:

$$\begin{aligned}
& + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} - \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \\
& - \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \\
& - \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \\
& + \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} - \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r}
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

$$\begin{array}{lcl}
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} & - & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} & - & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} & + & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} & - & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
- \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} & + & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
- \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} & + & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
- \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} & - & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
+ \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} & + & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} & + & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
- \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} & - & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} & - & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
+ \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} & - & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} & - & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} & + & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} & + & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} & + & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& -2 \cdot \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} + 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
& + 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} - 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
& + 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} - 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
& - 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} + 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u}
\end{aligned}$$

3.8 $i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$

$$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

monomer ci vectors:

$$\begin{aligned}
& + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \\
& + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \\
& - \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} - \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \\
& - \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} - \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r}
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
+ \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
+ \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
- \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
+ \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
- \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
- \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
- \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& +2 \cdot \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} - 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
& +2 \cdot \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} - 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
& +2 \cdot \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} - 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
& +2 \cdot \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} - 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}
\end{aligned}$$

3.9 $i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$

$$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}
\end{aligned}$$

monomer ci vectors:

$$\begin{aligned}
& + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} - \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \\
& + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} - \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \\
& - \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \\
& - \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r}
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

$$\begin{array}{lcl}
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} & - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} & + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
- & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} & - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} & - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} & - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} & + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
- & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} & - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} & + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
- & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} & - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} & - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} & - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} & + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
- & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} & - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} & - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& -2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \right| \\
& -2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \right| \\
& +2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \right| - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \right| \\
& +2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \right| - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \right|
\end{aligned}$$

3.10 $e^3 b_{2u} * e^3 b_{2u}$

$$e^3 b_{2u} * e^3 b_{2u}$$

$$+ \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right)$$

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \left| \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \right|
\end{aligned}$$

multiplied out dimer ci vectors:

[illegible]

summed up:

$$\begin{aligned}
& + \left| \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \right| + \left| \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g} \right|
\end{aligned}$$

3.11 $e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$

$$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$$

$$\begin{aligned}
& + \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) \\
& + \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right)
\end{aligned}$$

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
- \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
- \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
+ \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
- \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
- \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
+ \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
+ \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& +2 \cdot \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} - 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
& -2 \cdot \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} + 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
& -2 \cdot \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} + 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
& +2 \cdot \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} - 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}
\end{aligned}$$

3.12 $e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$

$$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

monomer ci vectors:

$$\begin{aligned}
& + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} - \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \\
& - \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \\
& + \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} - \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \\
& - \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r}
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
- \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
- \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
- \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
+ \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
- \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
+ \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
- \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
- \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& -2 \cdot \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} + 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
& + 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} - 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
& - 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} + 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
& + 2 \cdot \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} - 2 \cdot \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u}
\end{aligned}$$

3.13 $e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$

$$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

monomer ci vectors:

$$\begin{aligned}
& + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \\
& + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \\
& + \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \\
& + \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} + \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r}
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

$$\begin{array}{lcl}
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} & - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
- & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} & + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} & - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} & - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
- & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} & - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} & + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
- & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} & - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} & + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} & - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} & - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
- & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} & + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} & - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \right| \\
& +2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \right| \\
& -2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \right| \\
& -2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} \right|
\end{aligned}$$

3.14 $e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$

$$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^1(a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^1(a_u^r)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^0(a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^0(a_u^r)^1}_{b_{2u}^r} \right|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

$$\begin{array}{lcl}
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} & - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} & + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \\
- & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} & - \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} & - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} & - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} & + \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
- & \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} & - \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
- & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} & - \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \\
+ & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} & + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} \\
- & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}} & + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \\
- & \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} & + \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}} \\
- & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} & - \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
+ & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} & + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} \\
- & \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}} & + \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \\
- & \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} & + \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& -2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} \right| \\
& -2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \right| \\
& -2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} \right| + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} \right| \\
& -2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \right| + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} \right|
\end{aligned}$$

3.15 $e^3 b_{3u} * e^3 b_{3u}$

$$e^3 b_{3u} * e^3 b_{3u}$$

$$+ \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{array} \right)$$

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \left| \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \right|
\end{aligned}$$

multiplied out dimer ci vectors:

[illegible]

summed up:

$$\begin{aligned}
& + \left| \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \right| + \left| \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g} \right|
\end{aligned}$$

3.16 $e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$

$$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \updownarrow & \uparrow & \updownarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \updownarrow & \uparrow & \updownarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \updownarrow & \updownarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \updownarrow & \uparrow & \uparrow & \updownarrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \updownarrow & \uparrow & \uparrow & \updownarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \updownarrow & \updownarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \updownarrow & \uparrow & \updownarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \updownarrow & \uparrow & \updownarrow & \uparrow \end{pmatrix}
\end{aligned}$$

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

summed up:

$$+2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g} \right|$$

3.17 $e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$

$$e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right|$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r}
\end{aligned}$$

multiplied out dimer ci vectors:

summed up:

$$\begin{aligned}
& -4 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \right| \\
& + 4 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \right| \\
& - 4 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \right| \\
& + 4 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \right|
\end{aligned}$$

3.18 $i^3 b_{3u} * i^3 b_{3u}$

$$i^3 b_{3u} * i^3 b_{3u}$$

$$+ \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{array} \right)$$

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r} \right|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& + \left| \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+b_{2u} + b_{3g})^2(+b_{3u} + b_{2g})^1(+a_g + b_{1u})^1(+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^2(-b_{3u} + b_{2g})^1(-a_g + b_{1u})^1(-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+b_{2u} + b_{3g})^1(+b_{3u} + b_{2g})^2(+a_g + b_{1u})^0(+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-b_{2u} + b_{3g})^1(-b_{3u} + b_{2g})^2(-a_g + b_{1u})^0(-b_{1g} + a_u)^1}_{b_{3u}^r} \right|
\end{aligned}$$

multiplied out dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \\
- \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \\
+ \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \\
- \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}} \\
+ \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g}
\end{array}$$

summed up:

$$\begin{aligned}
& + \left| \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g} \right| + \left| \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g} \right|
\end{aligned}$$

4 Symmetry Conclusion

Dimer State	Symmetry
$S * Q + Q * S$	a_g
$S * Q - Q * S$	b_{1u}
$i^3 b_{2u} * i^3 b_{2u}$	a_g
$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$	a_g
$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$	b_{1u}
$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$	b_{1g}
$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$	a_u
$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$	b_{1g}
$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$	a_u
$e^3 b_{2u} * e^3 b_{2u}$	a_g
$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$	b_{1g}
$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$	a_u
$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$	b_{1g}
$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$	a_u
$e^3 b_{3u} * e^3 b_{3u}$	a_g
$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$	a_g
$e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$	b_{1u}
$i^3 b_{3u} * i^3 b_{3u}$	a_g

$\Rightarrow$ 7x a_g and 3x b_{1u} and 4x b_{1g} and 4x a_u

5 CI Vector Conclusion

Dimer States and Their CI Vectors:

- $S * Q + Q * S$:

$$+2 \cdot |2a2a0a0a| + 2 \cdot |2a2aa0a0| + 2 \cdot |2aa20aa0| + 2 \cdot |2aa2a00a| + 2 \cdot |a22a0aa0| + 2 \cdot |a22aa00a| \\ +2 \cdot |a2a20a0a| + 2 \cdot |a2a2a0a0|$$

- $S * Q - Q * S$:

$$-2 \cdot |2a2a0a0a| - 2 \cdot |2a2aa00a| - 2 \cdot |2aa20a0a| - 2 \cdot |2aa2a0a0| - 2 \cdot |a22a0a0a| - 2 \cdot |a22aa0a0| \\ -2 \cdot |a2a20aa0| - 2 \cdot |a2a2a00a|$$

- $i^3 b_{2u} * i^3 b_{2u}$:

$$+ |aa22aa00| + 2 \cdot |a22aa00a| - 2 \cdot |a22a0aa0| + 2 \cdot |a2a2a0a0| - 2 \cdot |a2a20a0a| - 2 \cdot |2a2aa0a0| \\ +2 \cdot |2a2a0a0a| - 2 \cdot |2aa2a00a| + 2 \cdot |2aa20aa0| + |22aa00aa|$$

- $i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$:

$$+2 \cdot |aa22aa00| - 2 \cdot |22aa00aa|$$

- $i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$:

$$+4 \cdot |a22aa0a0| - 4 \cdot |a22a0a0a| + 4 \cdot |a2a2a00a| - 4 \cdot |a2a20aa0| - 4 \cdot |2a2aa00a| + 4 \cdot |2a2a0aa0| \\ -4 \cdot |2aa2a0a0| + 4 \cdot |2aa20a0a|$$

- $i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$:

$$+2 \cdot |a22aaa00| - 2 \cdot |2aa2aa00| - 2 \cdot |aa22a00a| + 2 \cdot |aa220aa0| + 2 \cdot |22aa0aa0| - 2 \cdot |22aaa00a| \\ -2 \cdot |2aa200aa| + 2 \cdot |a22a00aa|$$

- $i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$:

$$-2 \cdot |a2a2aa00| + 2 \cdot |2a2aaa00| + 2 \cdot |aa22a0a0| - 2 \cdot |aa220a0a| + 2 \cdot |22aa0a0a| - 2 \cdot |22aaa0a0| \\ -2 \cdot |2a2a00aa| + 2 \cdot |a2a200aa|$$

- $i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$:

$$+2 \cdot |a22aaa00| - 2 \cdot |2aa2aa00| + 2 \cdot |aa22a00a| - 2 \cdot |aa220aa0| + 2 \cdot |22aa0aa0| - 2 \cdot |22aaa00a| \\ +2 \cdot |2aa200aa| - 2 \cdot |a22a00aa|$$

- $i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$:

$$-2 \cdot |a2a2aa00| + 2 \cdot |2a2aaa00| - 2 \cdot |aa22a0a0| + 2 \cdot |aa220a0a| + 2 \cdot |22aa0a0a| - 2 \cdot |22aaa0a0| \\ +2 \cdot |2a2a00aa| - 2 \cdot |a2a200aa|$$

- $e^3 b_{2u} * e^3 b_{2u}$:

$$+ |aa22aa00| - 2 \cdot |a22aa00a| + 2 \cdot |a22a0aa0| - 2 \cdot |a2a2a0a0| + 2 \cdot |a2a20a0a| + 2 \cdot |2a2aa0a0| \\ -2 \cdot |2a2a0a0a| + 2 \cdot |2aa2a00a| - 2 \cdot |2aa20aa0| + |22aa00aa|$$

- $e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$:
 $+2 \cdot |a22aaa00| - 2 \cdot |2aa2aa00| - 2 \cdot |aa22a00a| + 2 \cdot |aa220aa0| - 2 \cdot |22aa0aa0| + 2 \cdot |22aaa00a|$
 $+2 \cdot |2aa200aa| - 2 \cdot |a22a00aa|$

- $e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$:
 $-2 \cdot |a2a2aa00| + 2 \cdot |2a2aaa00| + 2 \cdot |aa22a0a0| - 2 \cdot |aa220a0a| - 2 \cdot |22aa0a0a| + 2 \cdot |22aaa0a0|$
 $+2 \cdot |2a2a00aa| - 2 \cdot |a2a200aa|$

- $e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$:
 $+2 \cdot |a22aaa00| - 2 \cdot |2aa2aa00| + 2 \cdot |aa22a00a| - 2 \cdot |aa220aa0| - 2 \cdot |22aa0aa0| + 2 \cdot |22aaa00a|$
 $-2 \cdot |2aa200aa| + 2 \cdot |a22a00aa|$

- $e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$:
 $-2 \cdot |a2a2aa00| + 2 \cdot |2a2aaa00| - 2 \cdot |aa22a0a0| + 2 \cdot |aa220a0a| - 2 \cdot |22aa0a0a| + 2 \cdot |22aaa0a0|$
 $-2 \cdot |2a2a00aa| + 2 \cdot |a2a200aa|$

- $e^3 b_{3u} * e^3 b_{3u}$:
 $+|22aaaa00| - 2 \cdot |2aa2a00a| + 2 \cdot |2aa20aa0| + 2 \cdot |2a2aa0a0| - 2 \cdot |2a2a0a0a| - 2 \cdot |a2a2a0a0|$
 $+2 \cdot |a2a20a0a| + 2 \cdot |a22aa00a| - 2 \cdot |a22a0aa0| + |aa2200aa|$

- $e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$:
 $+2 \cdot |22aaaa00| - 2 \cdot |aa2200aa|$

- $e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$:
 $-4 \cdot |2aa2a0a0| + 4 \cdot |2aa20a0a| + 4 \cdot |2a2aa00a| - 4 \cdot |2a2a0aa0| - 4 \cdot |a2a2a00a| + 4 \cdot |a2a20aa0|$
 $+4 \cdot |a22aa0a0| - 4 \cdot |a22a0a0a|$

- $i^3 b_{3u} * i^3 b_{3u}$:
 $+|22aaaa00| + 2 \cdot |2aa2a00a| - 2 \cdot |2aa20aa0| - 2 \cdot |2a2aa0a0| + 2 \cdot |2a2a0a0a| + 2 \cdot |a2a2a0a0|$
 $-2 \cdot |a2a20a0a| - 2 \cdot |a22aa00a| + 2 \cdot |a22a0aa0| + |aa2200aa|$